

Writing a Good Mathematical Paper

A compact guide for authors and AI assistants

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Write for a reader who knows the field but has not seen your argument. Let that reader identify the result, understand why the proof works, and recover every step. Spend words where they reduce mathematical difficulty.

1. Decide what the paper contributes. Choose a coherent principal result. Say what it establishes, why it matters, how it answers the question, and what remains open. Compare it with the strongest relevant prior result. Do not judge a contribution by theorem count, generality or conditional status alone. An abstract should name the actual result and mechanism.

2. Read the kind of mathematics you are writing. Read nearby human-authored papers in the same subject and genre. Study complete local arguments: terminology, hypothesis placement, sentence connections, notation, proof order, pace and attribution. Borrow the reason for a choice; write your own sentences. Check the edition and passage; “our main result” teaches little without the theorem that follows it.

3. Fix the statement before polishing its prose. Check domains, quantifiers, signs, endpoints, constants, parameter dependencies and the direction of every implication. State necessary assumptions where the claim is made; recall the relevant one when it explains a difficult inference. Check compressed restatements in titles, abstracts, captions and conclusions for a stronger claim. A sufficient condition is not an equivalence. Separate witnesses are not a common witness: over the natural numbers, $\forall n \exists x (x > n)$ does not imply $\exists x \forall n (x > n)$.

4. Organise by logical dependence. Give the reader the question, a useful example when needed, the result and the proof’s main idea before technical detail accumulates. State a lemma’s purpose before proving it. Put prerequisites before their use. Give subordinate results a clear place; cut repeated derivations. Choose length to suit the argument; there is no universal first-page proof or page quota.

5. Explain the hard transition. Name the quantity to control and the accuracy required before constructing an estimate. When a preliminary bound is too weak, identify what is missing and how the next step supplies it. Explain the target before the parameter schedule. For a small-integer contradiction, justify integrality, nonvanishing and $|N| < 1$ separately. Do not expand routine algebra while skipping the decisive inference. Make a proof paragraph advance a clear goal through an operation to what is now established or remains to prove.

6. Separate construction from completion. Account for every chosen coordinate, correction and error term; state what each choice preserves. Then justify convergence, interchange of limits and sums, attainment, or passage to an infinite object. Specify which parameters stay fixed and the order of successive limits. For example, $r_n = x - S_n$, $0 \leq r_n \leq \varepsilon_n$ and $\varepsilon_n \rightarrow 0$ give $S_n \rightarrow x$; the ability to choose another term alone does not. Distinguish pointwise convergence from the bound that permits interchanging a limit and a sum.

7. Make examples and figures do work. Use a small example to reveal a definition, choice, invariant, obstruction or difference between hypotheses. Show the operation and its consequence.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Say where the example stops. A finite check establishes an infinite claim only through a proved reduction. A diagram must identify its objects, maps, quantities and conventions; do not draw an arbitrary connected set as a smooth path without justification. Put the explanation beside the example it explains.

8. Use the field's words and purposeful notation. Keep established technical terms when their definitions fit: “moment sequences”, not “moment banks”. Replace private labels with the actual object or condition: write $u_n/2^n \rightarrow 0$ if that is what the adjective means; do not silently strengthen it to subexponential growth. Define symbols before use, keep their roles stable and remove notation that saves no repeated work. Distinguish a sum, its partial sums and its remainders.

9. Make sentences express the argument. Replace “We proceed with the estimate” by a mathematical reason: “Orthogonality makes the first sum zero; it remains to bound the second.” Use connectives to express actual reasons, consequences and conditions. Make referents clear. Use parallel syntax for parallel claims; keep a condition attached to what depends on it. Prefer verbs and named mathematical objects to stacked nouns. Remove throat-clearing, inflated significance and repeated summaries. Do not manufacture sentence variety or replace precise terms merely to avoid repetition.

10. Credit the exact input at its use. Distinguish applied theorems, adapted constructions, historical attribution, terminology and expository models. Check the original's hypotheses, version and passage. Credit supplied ideas and corrections locally. A stylistic analogy supplies no proof. Inspect the passage behind a search hit. Report only reading and checks actually performed.

11. State what has actually been established. State both logical scope and evidence: a conditional implication may have an ordinary proof or a formal proof; a computation or conjecture needs its own label. Identify cited inputs. Explain the argument on the page and retain its unproved inputs beside the result. A checked nearby lemma does not formalise the whole proof; missing formalisation does not invalidate an ordinary proof. A finite computation needs its scope and reproduction details. Claim novelty only after an appropriate literature comparison.

12. Preserve useful detail and informative failures. Let the short paper sustain its principal argument; let the longer record make the rest recoverable. Move detail only after checking the destination, then reconcile statements and evidence in both directions. For a failed route, keep the attempted implication, assumptions, witness and missing ingredient together. Distinguish a flawed step, a failed method and a counterexample to a theorem. Do not invent the history of discovery to motivate a proof.

13. Review changes as claims, then as prose. Compare old and new meanings, including words inside theorem statements. Removing a shorthand must not remove its hypotheses. Accept, repair or reject each substantive proposal for a stated reason. Generalise a lesson only as far as its evidence supports; keep its limit. Preserve useful unsuccessful revisions.

14. Read the delivered paper from a cold start. Recheck citations, labels, cross-reference destinations and consistency between paper forms. Render the final sources. Inspect formulas, captions, figures at reading size, page breaks and neighbouring pages. Read the argument without filling gaps from memory. Can the intended reader state the result, explain the hard step and identify its exact limit? Use observed reader difficulties to revise; do not substitute a successful build for that judgement.

Further reading. Halmos, *How to write mathematics*, §§3–5 (1970); Knuth, Larrabee and Roberts, *Mathematical Writing*, §1 (1988); Gowers, *Examples first!* (2007); Tao, *On writing*. Worked mathematical cases, sources and revision history: *Writing Mathematics from the Literature and Reviewed Revisions*.